1. $M = F \times$ perpendicular distance

$d = |2\cos(70) - 2\sin(70)| = 1.20$ in.

$M = 90(1.20) = 108$ lb in counterclockwise

2. $M = F \times d$

$M = 200(1.91) = 382$ N m clockwise

3. tan(a) = opposite over adjacent

$a = \arctan(0.601) = 31$ deg

4. R = sum of F = 116 kN downward

x = sum of M over R = 8.69 m

5. Rx = sum Fx Ry = sum Fy

R = sqrt(Rx2 plus Ry2) = 82.9 lb

theta = 465 deg above −x

M = 119 lb in CCW

6. Rx = sum Fx Ry = sum Fy

R = 164 lb at 723 deg below plus x

MB = sum of M = 498 lb in clockwise

7. F = 100 lb at 70 deg below plus x

sum of M about A = 0

xf = −1.70 ft